Jaccard Needs Exponentially Many Convex Scores

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Abstract

Jaccard similarity, also called intersection over union, and the F1 score are pointwise monotone transforms of one another. Their convex surrogate complexities are nevertheless exponentially different. For instance-wise multi-label prediction with s labels, we prove

$$2^{s-1} - 1 \ \leq \ \mathrm{CCdim}(L^{\mathrm{Jac}}) \ \leq \ 2^s - 1.$$

Thus every convex surrogate calibrated for Jaccard on all conditional label distributions needs exponentially many real-valued scores. In contrast, a recent tight result gives convex calibration dimension $\Theta(s^2)$ for F1. The lower bound comes from one explicit conditional distribution. It is supported on all sets containing a fixed core label, weights a set with d optional labels by $1/d!$, and makes exactly those 2^{s-1} reports Bayes optimal. Strict positive definiteness of the corresponding Jaccard block then collapses its feasible Bayes subspace to a point.

We also identify an algebraic phase transition. In the normalized-overlap family interpolating from Dice/F1 to Jaccard, the exact score-matrix rank is $s^2 - s + 2$ at the F1 endpoint but 2^s after any positive Jaccard correction. Finally, we reconcile our theorem with a NeurIPS 2024 claim that an s-score convex logistic surrogate is Bayes-consistent for every multi-label loss. A two-label rational example gives zero surrogate regret and Jaccard regret 1/10. The proof in that work transfers probability between two arbitrary label vectors, but its additive softmax probabilities form only an s-dimensional product family and cannot realize that transfer. Our results separate practical restricted-distribution methods from distribution-free convex calibration and show that, for Jaccard, this distinction is unavoidable.

1 Introduction

An image mask, tag vector, or structured binary prediction can be represented as a subset of s labels. Two standard scores for a report B and outcome A are

$$F_1(A, B) = \frac{2|A \cap B|}{|A| + |B|}, \qquad \mathrm{Jac}(A, B) = \frac{|A \cap B|}{|A \cup B|}. \tag{1}$$

Both equal one on identical nonempty sets, and we adopt the usual convention that both equal one on two empty sets. Pointwise they contain the same ordering information:

$$\mathrm{Jac}(A, B) = \frac{F_1(A, B)}{2 - F_1(A, B)}. \tag{2}$$

This close relationship makes the contrast in this paper surprising.

For a discrete loss, convex calibration dimension is the smallest d for which some loss convex in a score vector $u \in \mathbb{R}^d$, together with some link from scores to reports, is calibrated on every conditional outcome distribution (Ramaswamy & Agarwal, 2016). It asks how many real-valued outputs a distribution-free convex surrogate fundamentally needs. For F1, calibrated surrogates with $O(s^2)$ scores have been known since Zhang et al. (2020). A new exact analysis proves the matching lower bound $\mathrm{CCdim}(L^{F_1}) = \Theta(s^2)$ and identifies the exact score rank $s^2 - s + 2$ (Zhang, 2026). We prove that Jaccard instead needs $\Theta(2^s)$ scores.

This question matters because IoU is used precisely when coordinatewise metrics such as Hamming loss are inadequate. Bayes-optimal IoU decisions are already combinatorial even when conditional label probabilities factorize (Nowozin, 2014). Practical losses include the Lovász hinge and its extensions (Yu & Blaschko, 2020; Berman et al., 2018). Recent work shows that the usual Lovász-hinge link is inconsistent for Jaccard (Finocchiaro et al., 2025). That result concerns one surrogate and link. Our lower bound concerns every convex surrogate and every link below an exponential dimension.

There is an apparent contradiction. Mao et al. (2024) claim that a new s-score convex multi-label logistic loss is Bayes-consistent, with an excess-risk bound, for any multi-label loss. Jaccard is an explicit example in their theorem. Our lower bound rules this out. We resolve the contradiction constructively with an exact two-label counterexample and pinpoint the failed realizability step in their proof. The analogous full-output structured construction in Mao et al. (2023) has one free score for each label vector, so its probability-transfer argument does not have this problem and is consistent with our exponential upper bound.

Contributions.

1. We prove an exponential lower bound $\mathrm{CCdim}(L^{\mathrm{Jac}}) \geq 2^{s-1} - 1$. The standard affine upper bound is $2^s - 1$, determining the order exactly.
2. We give an explicit factorially weighted Bayes witness and a short combinatorial identity showing that precisely all reports containing one core label tie. This produces exponentially many independent active Bayes constraints at one distribution.
3. We prove a sharp rank discontinuity for a continuous family of normalized-overlap scores. The F1 endpoint has quadratic rank, while every positive Jaccard correction has full rank 2^s.
4. We give a rational two-label falsification certificate for the claimed general consistency of the low-dimensional multi-label logistic loss. It has zero surrogate regret but positive target regret, and does not depend on the empty-set convention.

The lower bound is about expected per-instance Jaccard loss, also called the decision-theoretic formulation. It does not concern a ratio formed after aggregating confusion counts across an entire population. We return to this scope distinction in Section 7.

2 Related work and position

Convex calibration dimension. Ramaswamy & Agarwal (2016) introduced convex calibration dimension for finite losses, proved affine and rank upper bounds, and developed the feasible-subspace lower bound used here. The flat-based elicitation framework of Finocchiaro et al. (2021) recovers that theorem for minimizable convex surrogates and gives stronger global tools. Neither work analyzes Jaccard or IoU. An important distinction in both is that matrix rank supplies a generic upper bound, not a lower bound for arbitrary convex surrogates. This is why our full-rank theorem and factorial Bayes witness are separate contributions.

F-measures. F1 Bayes decisions can be recovered from a quadratic collection of conditional statistics. Zhang et al. (2020) turn this structure into convex calibrated surrogates. Most recently, Zhang (2026) prove the exact rank and affine dimension and construct a feasible-subspace witness giving the matching $\Theta(s^2)$ calibration dimension. Our witness is inspired by the same general question but differs in both weighting and geometry. It creates 2^{s-1} independent active comparisons rather than a quadratic family.

Table 1: F1 and Jaccard are pointwise monotone-equivalent but have different convex surrogate geometry.

Quantity on s labels	F1/Dice	Jaccard/IoU
Score-matrix rank	$s^2 - s + 2$	2^s
Loss-column affine dimension	$s^2 - s + 1$	$2^s - 1$
Convex calibration dimension	$\Theta(s^2)$	$\Theta(2^s)$
Explicit bounds	$\Omega(s^2)$ to $s^2 - s + 1$	$2^{s-1} - 1$ to $2^s - 1$

IoU optimization and the Lovász hinge. Nowozin (2014) study Bayes decisions for IoU under a probabilistic model and derive an approximation for aggregate semantic segmentation. The Lovász hinge and Lovász-Softmax use submodularity to obtain practical convex objectives (Yu & Blaschko, 2020; Berman et al., 2018). For symmetric submodular losses, Finocchiaro et al. (2022) identify the structured abstain target that the Lovász hinge actually embeds. The recent label-dependent extension proves that the usual sign link is inconsistent for Jaccard when $s \geq 3$ (Finocchiaro et al., 2025). It leaves open whether a different convex surrogate or link of comparable dimension could be calibrated. Theorem 2 rules out all of them.

Low-dimensional structured surrogates. The full-output structured logistic loss of Mao et al. (2023) has a free score for each of the 2^s label vectors. The multi-label specialization of Mao et al. (2024) replaces those scores by s additive label scores and claims the same guarantee for every target loss. Section 6 shows exactly why that reduction fails for Jaccard. Partial calibration under low-noise conditions and aggregation of several low-dimensional problems are studied by Nueve et al. (2024). Those results complement our full-simplex lower bound.

Table 1 summarizes the sharp separation. The F1 entries are from Zhang (2026); the Jaccard entries are proved here.

3 Setup and calibration dimension

Let $[s] = \{1, \dots, s\}$ and let outcomes and reports both range over $\mathcal{Y} = 2^{[s]}$, of cardinality $N = 2^s$. Define the Jaccard score matrix $K \in \mathbb{R}^{N \times N}$ and loss matrix L^{Jac} by

$$K_{A,B} = \begin{cases} 1, & A = B = \varnothing, \\ |A \cap B|/|A \cup B|, & \text{otherwise}, \end{cases} \qquad L^{\mathrm{Jac}} = \mathbf{1}\mathbf{1}^\top - K. \tag{3}$$

Rows index outcomes and columns index reports. For a conditional distribution $q \in \Delta_N$, the Bayes reports maximize $q^\top K_{\cdot,B}$.

A surrogate is a function $\psi : \mathcal{C} \to \mathbb{R}_+^N$, where $\mathcal{C} \subseteq \mathbb{R}^d$ is convex and each coordinate ψ_y is convex. It is calibrated for L if it admits a link from $\mathcal{C}$ to reports such that, at every $q \in \Delta_N$, approaching optimal surrogate risk cannot retain a suboptimal target report. The convex calibration dimension $\mathrm{CCdim}(L)$ is the smallest such d (Ramaswamy & Agarwal, 2016). This definition permits arbitrary convex losses and arbitrary links.

For report B, its trigger polytope is

$$Q_B = \{q \in \Delta_N : q^\top L_{\cdot,B} \leq q^\top L_{\cdot,C} \text{ for every } C\}. \tag{4}$$

Let $\mu_{Q_B}(q)$ be the dimension of the largest linear subspace of two-sided feasible directions of Q_B at q. The feasible-subspace theorem of Ramaswamy & Agarwal (2016) states

$$\mathrm{CCdim}(L) \geq \|q\|_0 - \mu_{Q_B}(q) - 1, \qquad q \in Q_B. \tag{5}$$

The newer flat-based framework recovers this bound and can strengthen it in other problems (Finocchiaro et al., 2021). Our witness makes $\mu_{Q_B}(q) = 0$, so the original theorem is already sharp enough.

4 A full-rank phase transition

We first isolate an algebraic distinction between F1 and Jaccard. For $0 \le \lambda < 2$, define the normalized-overlap score

$$S_\lambda(A,B) = \frac{(2-\lambda)|A \cap B|}{|A| + |B| - \lambda|A \cap B|} \tag{6}$$

for nonempty pairs, with value zero if exactly one set is empty and value one if both are empty. Then $S_0 = F_1$ and $S_1 = \text{Jac}$. If $D = F_1(A,B)$, then

$$S_\lambda(A,B) = \frac{(2-\lambda)D}{2-\lambda D}, \tag{7}$$

so every member is a pointwise strictly increasing transform of F1.

Theorem 1 (Rank phase transition)**.** *Under the empty-empty convention above, the score matrix of S_0 has rank $s^2 - s + 2$. For every $\lambda \in (0,2)$, the score matrix of S_λ is strictly positive definite and has rank 2^s.*

The endpoint rank is the exact F1 result of Zhang (2026). We prove the positive-λ statement. Write $x = |A \cap B|$, $a = |A|$, and $b = |B|$. For nonempty sets,

$$S_\lambda(A,B) = (2-\lambda)\int_0^\infty x e^{-t(a+b-\lambda x)}\,dt. \tag{8}$$

Let $\chi_A \in \{0,1\}^s$ be the incidence vector. Expanding $xe^{\lambda t x}$ into tensor powers gives, for coefficients c_A on the nonempty sets,

$$c^\top S_\lambda c = (2-\lambda)\int_0^\infty \sum_{r\ge1} \frac{(\lambda t)^{r-1}}{(r-1)!} \left\| \sum_{A\ne\varnothing} c_A e^{-t|A|} \chi_A^{\otimes r} \right\|_2^2 dt. \tag{9}$$

This is nonnegative. If it vanished, every displayed tensor sum would vanish. For any nonempty $T \subseteq [s]$, inspect a coordinate of the $|T|$th tensor that lists every element of T. It gives

$$\sum_{A \supseteq T} c_A e^{-t|A|} = 0. \tag{10}$$

At any fixed $t > 0$, these are all coordinates of the Boolean upper-zeta transform of $c_A e^{-t|A|}$. The transform is invertible, so every $c_A = 0$. The nonempty block is strictly positive definite. The empty-set convention adds an isolated 1×1 block, proving the theorem. A different positive definiteness proof for the Jaccard endpoint was given by Bouchard et al. (2013). Equation (9) is included because it covers the full family and makes the discontinuity transparent.

Full score rank alone is not a lower bound on arbitrary convex calibration dimension. It does imply that the N score columns, and hence the N loss columns, are affinely independent. Therefore

$$\operatorname{affdim}(L^{\text{Jac}}) = N - 1. \tag{11}$$

The generic affine construction of Ramaswamy & Agarwal (2016) now gives $\operatorname{CCdim}(L^{\text{Jac}}) \le N - 1$. The next section proves that no polynomial improvement is possible.

5 The exponential calibration barrier

Fix one core label, denoted $\star$, and write $O = [s] \setminus \{\star\}$, with $n = s - 1$. Consider the 2^n outcomes containing the core:

$$\mathcal{U} = \{\{\star\} \cup D : D \subseteq O\}. \tag{12}$$

Define a distribution p supported on $\mathcal{U}$ by

$$p_{\{\star\}\cup D} = \frac{1}{Z_n |D|!}, \qquad Z_n = \sum_{d=0}^{n} \binom{n}{d} \frac{1}{d!}. \tag{13}$$

Lemma 1 (Factorial tie)**.** *Under p, precisely the reports in $\mathcal{U}$ maximize expected Jaccard similarity.*

Here is the main identity. For $C \subseteq O$, ignore the common normalizer Z_n and write the expected score of report $\{\star\} \cup C$ as

$$W(C) = \sum_{D \subseteq O} \frac{1}{|D|!} \frac{1 + |C \cap D|}{1 + |C \cup D|}. \tag{14}$$

Then

$$W(C) = \sum_{d=0}^{n} \binom{n}{d} \frac{1}{(d+1)!}, \tag{15}$$

independent of C. To see why, add one optional label $j \notin C$ and pair $D = E$ with $D = E \cup \{j\}$. Put $a = |E \cap C|$, $b = |E \setminus C|$, and $c = |C|$. The paired increment is

$$\frac{1}{c+b+2} \left[\frac{1}{(a+b+1)!} - \frac{a+1}{(a+b)!(c+b+1)} \right]. \tag{16}$$

For every fixed b, summing over the a choices cancels by

$$\sum_{a=0}^{c} \binom{c}{a} \frac{1}{(a+b+1)!} = \frac{1}{c+b+1} \sum_{a=0}^{c} \binom{c}{a} \frac{a+1}{(a+b)!}. \tag{17}$$

Thus adding j does not change W, and $W(C) = W(\varnothing)$ gives (15). If a report omits $\star$, adding $\star$ strictly increases its intersection with every outcome in the support while leaving the union unchanged. Such a report is strictly suboptimal. Appendix C gives the complete cancellation proof.

Theorem 2 (Exponential Jaccard calibration dimension)**.** *For every $s \geq 2$,*

$$2^{s-1} - 1 \leq \mathrm{CCdim}(L^{\mathrm{Jac}}) \leq 2^s - 1. \tag{18}$$

In particular, $\mathrm{CCdim}(L^{\mathrm{Jac}}) = \Theta(2^s)$*.*

Proof. The upper bound follows from (11). For the lower bound, let $r = |\mathcal{U}| = 2^{s-1}$ and fix $B_0 \in \mathcal{U}$. By Lemma 1, $p \in Q_{B_0}$ and every comparison between B_0 and $B \in \mathcal{U}$ is active.

Restrict the score matrix to rows and columns in $\mathcal{U}$. It is a principal submatrix of the strictly positive definite Jaccard matrix, so its r columns are linearly independent. Therefore the $r-1$ differences

$$K_{\mathcal{U},B} - K_{\mathcal{U},B_0}, \qquad B \in \mathcal{U} \setminus \{B_0\}, \tag{19}$$

are linearly independent. Their common orthogonal complement in $\mathbb{R}^r$ is one-dimensional. The vector $p_{\mathcal{U}}$ belongs to it by the Bayes tie, so that complement is $\mathrm{span}(p_{\mathcal{U}})$.

Now let v be a two-sided feasible direction of Q_{B_0} at p. Since p is zero outside $\mathcal{U}$, two-sided feasibility in the simplex forces v to be zero there. Every active comparison forces $v_{\mathcal{U}}$ to be orthogonal to (19). Hence $v_{\mathcal{U}} = \alpha p_{\mathcal{U}}$. The simplex constraint gives $0 = \mathbf{1}^\top v = \alpha$, because $\mathbf{1}^\top p = 1$. Thus $v = 0$ and $\mu_{Q_{B_0}}(p) = 0$. Since $\|p\|_0 = r$, (5) gives $\mathrm{CCdim}(L^{\mathrm{Jac}}) \geq r - 1$. $\square$

The proof uses positive definiteness only to certify independence of active Bayes comparisons. Positive definiteness by itself would give only an affine upper bound. The factorial witness is what turns the algebra into a lower bound for every convex surrogate.

6 A two-label falsification certificate

We now reconcile Theorem 2 with the purported s-score surrogate of Mao et al. (2024). Their general multi-label logistic loss can be written

$$\widetilde{L}(h, y) = \sum_{B \subseteq [s]} \mathrm{Jac}(B, y) \log\left(\sum_{C \subseteq [s]} e^{\mathrm{score}_h(C) - \mathrm{score}_h(B)} \right), \quad \mathrm{score}_h(B) = \sum_{i=1}^{s} y_i^B h_i, \tag{20}$$

where $y_i^B = 1$ for $i \in B$ and -1 otherwise. The link predicts $\{i : h_i \geq 0\}$. Theorem 5.1 and Corollary 5.2 of that work claim a Jaccard excess-risk bound and Bayes consistency.

Take $s = 2$, order sets as $\varnothing, \{1\}, \{2\}, \{1, 2\}$, and put

$$p = \left(0, \frac{3}{5}, \frac{1}{10}, \frac{3}{10}\right). \tag{21}$$

The expected Jaccard scores of the four reports are

$$a = \mathbb{E}_p[K_{Y,\cdot}] = \left(0, \frac{3}{4}, \frac{1}{4}, \frac{13}{20}\right). \tag{22}$$

Thus the unique Bayes report is $\{1\}$.

Let

$$s_B(h) = \frac{e^{\mathrm{score}_h(B)}}{\sum_C e^{\mathrm{score}_h(C)}}.$$

The conditional surrogate risk from (20) is exactly

$$-\sum_B a_B \log s_B(h). \tag{23}$$

The probabilities $s_B(h)$ form a product distribution on the two labels. The information projection of $a / \sum_B a_B$ onto this family matches its label marginals, which here are

$$m_1 = \frac{a_{\{1\}} + a_{\{1,2\}}}{\sum_B a_B} = \frac{28}{33}, \qquad m_2 = \frac{a_{\{2\}} + a_{\{1,2\}}}{\sum_B a_B} = \frac{6}{11}. \tag{24}$$

Both exceed 1/2. Consequently the unique finite minimizer is

$$h_1^* = \frac{1}{2} \log \frac{28}{5} > 0, \qquad h_2^* = \frac{1}{2} \log \frac{6}{5} > 0. \tag{25}$$

The link predicts $\{1, 2\}$, whose Jaccard score is 13/20, instead of the Bayes score 3/4. Its target regret is 1/10 at zero surrogate regret. The distribution is supported on nonempty outcomes, so this contradiction is independent of how $\mathrm{Jac}(\varnothing, \varnothing)$ is defined.

The proof gap is equally concrete. The proof of Mao et al. (2024) defines s^μ by moving probability between two arbitrary label vectors while holding every other coordinate fixed, then asserts that completeness of the s scalar score classes realizes s^μ. But additive softmax probabilities lie on an s-dimensional product manifold. With two labels they always satisfy

$$s_\varnothing s_{\{1,2\}} = s_{\{1\}} s_{\{2\}}. \tag{26}$$

An arbitrary two-coordinate transfer does not preserve this identity. Coordinatewise completeness makes every h_i attainable, not every point of the $(2^s - 1)$-dimensional probability simplex. The same transfer appears in a recent EUM analysis (Mohri & Zhong, 2026), but that work addresses population-level generalized metrics rather than the per-instance loss of Theorem 2.

7 Implications and limitations

Why monotone equivalence is insufficient. Equation (2) preserves the ranking of scores for a fixed outcome, but Bayes decisions rank expectations of those scores. Expectation does not commute with a nonlinear monotone transform. The factorial witness shows that this distinction changes convex calibration dimension from quadratic for F1 to exponential for Jaccard. The rank transition in Theorem 1 is an algebraic reflection of the same phenomenon.

What the result rules out. No loss convex in a subexponential-dimensional report vector, with any link, can be calibrated for expected instance-wise Jaccard on the full conditional simplex. This includes more than coordinate-separable or additive-score losses. The obstruction is distribution-free and link-independent.

What remains possible. The theorem does not rule out nonconvex surrogates, restricted conditional families, partial calibration under low noise, or several parallel low-dimensional problems whose total information is exponential. These are genuine escape routes (Nueve et al., 2024). Nor is calibration dimension itself a running-time lower bound. A method may represent exponentially many scores implicitly and use structured inference. Full-output structured surrogates (Mao et al., 2023) and the generic affine construction match the exponential order.

The target here is the decision-theoretic risk $\mathbb{E}[1 - \mathrm{Jac}(h(X), Y)]$. Dataset-level micro IoU, ratios of expected aggregate counts, and empirical utility maximization define different decision problems (Nowozin, 2014; Mohri & Zhong, 2026). Our theorem does not transfer to those formulations without a separate analysis.

Finally, our upper and lower bounds differ by a factor of two. Determining the exact Jaccard calibration dimension is open. It is also open whether every positive member of (6) has exponential calibration dimension. The full-rank proof alone cannot establish that claim, and we do not make it.

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A Calibration-dimension details

This appendix records the exact geometric objects used in Theorem 2. Let $L \in \mathbb{R}_+^{n\times k}$ be a finite loss matrix, with outcome distributions in Δ_n. For report t, write

$$Q_t^L = \{q \in \Delta_n : q^\top L_{\cdot,t} \le q^\top L_{\cdot,t'} \text{ for every } t' \in [k]\}. \tag{27}$$

For a convex set $Q \subseteq \mathbb{R}^n$ and $p \in Q$, its cone of feasible directions is

$$F_Q(p) = \{v : \text{there is } \epsilon_0 > 0 \text{ such that } p + \epsilon v \in Q \text{ for every } \epsilon \in (0, \epsilon_0)\}.$$

The feasible subspace is $F_Q(p) \cap (-F_Q(p))$, and its dimension is $\mu_Q(p)$. Theorem 16 of Ramaswamy & Agarwal (2016) gives

$$\mathrm{CCdim}(L) \ge \|p\|_0 - \mu_{Q_t^L}(p) - 1 \quad \text{for every } p \in Q_t^L. \tag{28}$$

It applies to the calibration definition used in their paper and does not require the conditional surrogate risk to attain its infimum.

For completeness, their affine upper bound states

$$\mathrm{CCdim}(L) \le \mathrm{affdim}(L), \tag{29}$$

where $\mathrm{affdim}(L)$ is the affine dimension of the set of loss columns. One way to view the construction is to choose affine coordinates for the loss columns, estimate the corresponding conditional expectations with a convex quadratic loss, and decode by minimizing the reconstructed target risk.

B Proof details for the rank transition

We prove the positive-λ part of Theorem 1 with all interchanges explicit. Let $\mathcal{Y}_+ = 2^{[s]} \setminus \{\varnothing\}$. For $A, B \in \mathcal{Y}_+$, abbreviate

$$x = |A \cap B|, \qquad a = |A|, \qquad b = |B|.$$

Because $0 \le \lambda < 2$ and $x \le \min(a, b)$, the denominator in (6) is positive. The elementary identity

$$\frac{x}{a + b - \lambda x} = \int_0^\infty x e^{-t(a+b-\lambda x)} dt \tag{30}$$

gives (8).

For incidence vectors $\chi_A, \chi_B \in \{0,1\}^s$,

$$x^r = \langle \chi_A^{\otimes r}, \chi_B^{\otimes r} \rangle. \tag{31}$$

Moreover,

$$xe^{\lambda t x} = \sum_{r=1}^{\infty} \frac{(\lambda t)^{r-1}}{(r-1)!} x^r. \tag{32}$$

All coefficients in this series are nonnegative. Since the set family is finite, Tonelli's theorem applies after separating the positive and negative parts of the finite coefficient vector, or equivalently after first expanding the finite quadratic form. Equations (30)–(32) yield

$$\sum_{A,B \in \mathcal{Y}_+} c_A c_B S_\lambda(A,B) = (2-\lambda) \int_0^\infty \sum_{r=1}^{\infty} \frac{(\lambda t)^{r-1}}{(r-1)!} \left\langle \sum_A c_A e^{-t|A|} \chi_A^{\otimes r}, \sum_B c_B e^{-t|B|} \chi_B^{\otimes r} \right\rangle dt \tag{1}$$

$$= (2-\lambda) \int_0^\infty \sum_{r=1}^{\infty} \frac{(\lambda t)^{r-1}}{(r-1)!} \left\| \sum_A c_A e^{-t|A|} \chi_A^{\otimes r} \right\|_2^2 dt. \tag{33}$$

The integrand is continuous and nonnegative. If the integral is zero, the integrand is zero for every $t > 0$. Since $\lambda > 0$, every tensor norm in the sum is zero.

Fix a nonempty $T = \{i_1, \ldots, i_r\} \subseteq [s]$. The $(i_1, \ldots, i_r)$ coordinate of the rth tensor sum is

$$\sum_{A \in \mathcal{Y}_+} c_A e^{-t|A|} \prod_{j=1}^{r} \chi_A(i_j) = \sum_{A \supseteq T} c_A e^{-t|A|}. \tag{34}$$

Fix any $t > 0$ and put $d_A = c_A e^{-t|A|}$. Equation (34) says that the upper-zeta transform $T \mapsto \sum_{A \supseteq T} d_A$ is zero on the poset of nonempty subsets. Ordered by decreasing cardinality, its matrix is triangular with ones on the diagonal. Hence $d_A = 0$ for every A, and therefore $c_A = 0$. The nonempty block is strictly positive definite. The empty set has score zero against every nonempty set and score one against itself, so it contributes a separate positive 1×1 block.

At $\lambda = 0$, only the $r = 1$ term remains in (32). The variation of $e^{-t|A|}$ with cardinality allows more than s rank, but not full rank. Zhang (2026) prove the exact value $s^2 - s + 2$, including the empty-set block.

Finally, if a square matrix K of size N is nonsingular, its N columns are linearly independent and thus affinely independent. The affine map $k_B \mapsto \mathbf{1} - k_B$ preserves affine independence, so the loss columns of $\mathbf{1}\mathbf{1}^\top - K$ have affine dimension $N-1$. This proves (11).

C The factorial tie identity

We give a full proof of Lemma 1. Recall that O contains the $n = s-1$ optional labels. For $C \subseteq O$, define $W(C)$ by (14). We first show that $W(C)$ is invariant under adding an element.

Fix $j \in O \setminus C$. Pair the outcome index $D = E$ with $D = E \cup \{j\}$ for each $E \subseteq O \setminus \{j\}$. Put

$$a = |E \cap C|, \qquad b = |E \setminus C|, \qquad c = |C|.$$

For $D = E$, adding j to the report leaves the intersection unchanged and increases the union size by one. Its contribution to $W(C \cup \{j\}) - W(C)$ is

$$-\frac{a+1}{(a+b)!(c+b+1)(c+b+2)}. \tag{35}$$

For $D = E \cup \{j\}$, adding j to the report leaves the union unchanged and increases the intersection by one. Its contribution is

$$\frac{1}{(a+b+1)!(c+b+2)}. \tag{36}$$

Their sum is (16).

For a fixed b, the number of choices of E having a given a is $\binom{c}{a}\binom{n-c-1}{b}$. The second binomial coefficient and the factor $1/(c+b+2)$ do not depend on a. It remains to prove (17). Multiply its left-hand side by $c+b+1$ and use $c+b+1 = (c-a)+(a+b+1)$:

$$(c+b+1)\sum_{a=0}^{c}\binom{c}{a}\frac{1}{(a+b+1)!} = \sum_{a=0}^{c}\binom{c}{a}\frac{1}{(a+b)!} + \sum_{a=0}^{c}(c-a)\binom{c}{a}\frac{1}{(a+b+1)!} \tag{2}$$

$$= \sum_{a=0}^{c}\binom{c}{a}\frac{1}{(a+b)!} + c\sum_{a=0}^{c-1}\binom{c-1}{a}\frac{1}{(a+b+1)!} \tag{3}$$

$$= \sum_{a=0}^{c}\binom{c}{a}\frac{1}{(a+b)!} + \sum_{a=0}^{c}a\binom{c}{a}\frac{1}{(a+b)!} \tag{4}$$

$$= \sum_{a=0}^{c}(a+1)\binom{c}{a}\frac{1}{(a+b)!}. \tag{37}$$

In the third line we shifted the index and used $a\binom{c}{a} = c\binom{c-1}{a-1}$. Hence every fixed-$b$ sum of paired increments is zero, and $W(C\cup\{j\}) = W(C)$.

Repeatedly removing elements from C gives $W(C) = W(\varnothing)$. Directly,

$$W(\varnothing) = \sum_{D\subseteq O}\frac{1}{|D|!(1+|D|)} = \sum_{d=0}^{n}\binom{n}{d}\frac{1}{(d+1)!}, \tag{38}$$

which proves (15). After division by Z_n, all reports in $\mathcal{U}$ therefore tie.

Now take any report B with $\star \notin B$ and put $B^+ = B\cup\{\star\}$. Every outcome A in the support of p contains $\star$. Therefore

$$|B^+\cup A| = |B\cup A|, \qquad |B^+\cap A| = |B\cap A|+1.$$

The Jaccard score of B^+ is strictly larger for every such A. Since all support probabilities are positive, B is strictly suboptimal. Thus exactly $\mathcal{U}$ is Bayes optimal.

D Feasible-subspace calculation

We expand the geometric step in Theorem 2. Let $K_{\mathcal{U},\mathcal{U}}$ be the $r\times r$ principal Jaccard block, where $r = 2^{s-1}$. It is positive definite by Theorem 1. Fix $B_0 \in \mathcal{U}$ and form the matrix whose $r-1$ columns are

$$d_B = K_{\mathcal{U},B} - K_{\mathcal{U},B_0}, \qquad B\in\mathcal{U}\setminus\{B_0\}. \tag{39}$$

These columns are independent. Indeed, if $\sum_{B\neq B_0}\alpha_B d_B = 0$, define coefficients $\beta_B = \alpha_B$ for $B\neq B_0$ and $\beta_{B_0} = -\sum_{B\neq B_0}\alpha_B$. Then $K_{\mathcal{U},\mathcal{U}}\beta = 0$. Nonsingularity gives $\beta = 0$, hence every $\alpha_B = 0$. Thus the matrix D of differences has rank $r-1$.

The tie identity says $D^\top p_{\mathcal{U}} = 0$. Since $\dim\ker(D^\top) = 1$ and $p_{\mathcal{U}} \neq 0$,

$$\ker(D^\top) = \operatorname{span}(p_{\mathcal{U}}). \tag{40}$$

Let $v \in F_{Q_{B_0}}(p)\cap(-F_{Q_{B_0}}(p))$. If $A\notin\mathcal{U}$, then $p_A = 0$. Feasibility of both v and $-v$ in the nonnegative orthant forces $v_A = 0$. For every $B\in\mathcal{U}$, the target risks of B and B_0 tie at p. The corresponding inequality in Q_{B_0} is active. Feasibility in both directions forces its directional derivative to be zero:

$$v_{\mathcal{U}}^\top(L^{\mathrm{Jac}}_{\mathcal{U},B} - L^{\mathrm{Jac}}_{\mathcal{U},B_0}) = 0. \tag{41}$$

Since loss differences are negative score differences, (41) is $D^\top v_{\mathcal{U}} = 0$. By (40), $v_{\mathcal{U}} = \alpha p_{\mathcal{U}}$. Finally, two-sided feasibility in the affine hull of the simplex requires $\mathbf{1}^\top v = 0$. Because $\mathbf{1}^\top p = 1$, this implies $\alpha = 0$. The feasible subspace is $\{0\}$ and has dimension zero.

Notice that constraints against reports outside $\mathcal{U}$ were not needed. They are strict at p by Lemma 1 and therefore impose no additional equality on a two-sided feasible direction.

E Exact analysis of the logistic counterexample

For the set order $\varnothing, \{1\}, \{2\}, \{1,2\}$, the Jaccard score matrix is

$$K = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/2 \\ 0 & 0 & 1 & 1/2 \\ 0 & 1/2 & 1/2 & 1 \end{pmatrix}. \tag{42}$$

Multiplying (21) by this matrix gives (22). The unique maximum 3/4 occurs at report $\{1\}$.

We next derive the conditional surrogate risk. Let $r_B = \mathrm{score}_h(B)$. Each term in (20) satisfies

$$\log \sum_C e^{r_C - r_B} = \log \sum_C e^{r_C} - r_B = -\log s_B(h). \tag{43}$$

Taking expectation over Y and interchanging two finite sums gives

$$\mathbb{E}_p \widetilde{L}(h, Y) = -\sum_B \mathbb{E}_p[\mathrm{Jac}(B, Y)] \log s_B(h) = -\sum_B a_B \log s_B(h). \tag{44}$$

Because $r_B = \sum_i y_i^B h_i$, $s_B(h)$ is the product of two Bernoulli probabilities. If $q_B = a_B/A$ with $A = \sum_B a_B = 33/20$, then (44) is A times the cross entropy from q to this product family. Writing $\pi_i = \Pr_{s(h)}(i \in B)$, the part depending on h is

$$A \sum_{i=1}^{2} [-m_i \log \pi_i - (1 - m_i) \log(1 - \pi_i)], \tag{45}$$

up to a constant independent of h. Each binary cross entropy is strictly convex in its logit and uniquely minimized at $\pi_i = m_i$. Direct arithmetic gives

$$A = \frac{33}{20}, \qquad m_1 = \frac{3/4 + 13/20}{33/20} = \frac{28}{33}, \qquad m_2 = \frac{1/4 + 13/20}{33/20} = \frac{6}{11}. \tag{46}$$

Under the $\{-1, 1\}$ score encoding, $\pi_i = e^{h_i}/(e^{h_i} + e^{-h_i})$. Solving for h_i yields (25). Both coordinates are positive, so the sign link returns $\{1, 2\}$. Its target loss exceeds the Bayes loss by

$$\left(1 - \frac{13}{20}\right) - \left(1 - \frac{3}{4}\right) = \frac{1}{10}. \tag{47}$$

The conditional surrogate risk is minimized exactly, so its regret is zero.

For reference, the product constraint (26) follows immediately from the four unnormalized weights

$$e^{-h_1 - h_2}, \quad e^{h_1 - h_2}, \quad e^{-h_1 + h_2}, \quad e^{h_1 + h_2}.$$

Both cross-products equal one before applying the common softmax normalizer.

F Finite exact checks

The proof is symbolic, but small cases provide useful falsification checks. For $s = 2, 3, 4, 5$, exact rational calculations for the witness (13) give

s	$\lvert\mathcal{U}\rvert$	tied score	best score outside $\mathcal{U}$	$\mathrm{rank}(K_{\mathcal{U},\mathcal{U}})$
2	2	3/4	1/4	2
3	4	13/21	2/7	4
4	8	73/136	39/136	8
5	16	501/1045	292/1045	16

(48)

The calculation enumerates all 2^s outcomes and reports using rational arithmetic, checks equality of the scores over $\mathcal{U}$, strict separation outside $\mathcal{U}$, and exact rank of the rational principal matrix. The two-label certificate separately checks

$$a = (0, 3/4, 1/4, 13/20), \quad (m_1, m_2) = (28/33, 6/11), \quad \text{regret} = 1/10.$$

No numerical approximation is used in these checks.